

Nishanth Gaikoti

Data Engineer II

nishanthg.jobs@gmail.com • 937-931-1112 • [LinkedIn](#)

SUMMARY

Data Engineer with 6+ years architecting Snowflake-native data lakehouse platforms and cloud data warehousing solutions across healthcare, insurance, and analytics domains. Expert in data pipeline architecture, ELT automation (dbt, Airflow, Fivetran), dimensional modeling, and stream processing. Proven track record implementing DataOps practices, Infrastructure as Code, and data quality frameworks to deliver scalable, cost-optimized solutions. Strong collaborator skilled in requirements gathering and stakeholder communication.

PROFESSIONAL EXPERIENCE

Data Engineer II

OPTUM (UnitedHealth Group)

Apr 2025 – Present | Remote, USA

- Architected cloud-native data pipeline architecture using Python and Apache Airflow on Snowflake, orchestrating 15 daily batch and stream processing workflows that ingest 2.5M healthcare records from Epic EHR via RESTful API integration, compressing data availability from 24 to 4 hours
- Developed SQL transformation models using dbt to standardize 40+ HEDIS quality metrics with built-in data validation and automated profiling, improving calculation accuracy from 95% to 98% while supporting clinical data analysis
- Built dimensional star schema models in Snowflake for provider performance and patient outcomes, implementing SCD Type 2 logic for historical tracking through collaborative requirements gathering with analytical teams
- Engineered event-driven Kafka Connect pipeline processing 50K/hour real-time claims events into Snowflake via Snowpipe for fraud detection dashboards with 5 minute latency, utilizing JSON processing and schema evolution
- Optimized Snowflake warehouse through performance tuning, query optimization, clustering strategies, and materialized views, reducing execution times 35% and achieving \$12K monthly cost savings through automated resource management
- Led Agile sprint planning and stand-ups as squad lead, implementing Great Expectations framework with data contracts and observability monitoring that reduced production incidents 60% through proactive troubleshooting while maintaining HIPAA compliance

Data Engineer

Mercury Insurance

May 2023 – Feb 2025 | McKinney, TX

- Architected scalable data lakehouse platform using Apache Airflow and AWS Glue, processing 17M daily customer records from 12 source systems into S3-based medallion architecture (bronze/silver/gold) with Parquet, Avro, and ORC formats, then loading into Redshift via serverless Lambda functions
- Built transformation layer using PySpark on AWS EMR with Docker container orchestration (ECS) to denormalize policy hierarchies across PostgreSQL, Oracle, and MS-SQL sources, processing 2TB monthly with shell scripting and XML/JSON processing
- Implemented Change Data Capture using Debezium and Kafka Streams for real-time policy updates into S3 data lake with Protocol Buffers serialization, reducing data freshness from T+1 to 15 minutes with automated data quality checks
- Engineered dimensional models in Redshift with SCD Type 2 temporal logic using distribution keys and partitioning strategies for database optimization, powering 40+ Amazon QuickSight dashboards for business intelligence
- Automated Infrastructure as Code using Terraform and GitHub Actions CI/CD pipelines with pytest unit testing and code review workflows, reducing deployment cycles from 3 hours to 20 minutes while maintaining Git version control
- Established data governance using AWS Glue Data Catalog for metadata management, implementing Master Data Management framework with column-level data lineage tracking and 200+ documented assets for regulatory compliance
- Partnered with actuarial teams in bi-weekly Scrum retrospectives for wildfire risk modeling datasets, orchestrating Hive-to-Parquet ETL jobs processing 4M+ geospatially-enriched records that compressed data science prep time 70%

Performance Data Engineer

FLUTTER ENTERTAINMENT

Jun 2018 – Jul 2021 | Hyderabad, India

- Built production-grade ETL pipelines using Apache Airflow processing 500K daily betting transactions from PostgreSQL into AWS Redshift, supporting 200+ operations staff with incident management and production support
- Designed star schema models for customer service metrics (AHT, FCR, CSAT) and fraud detection with indexing strategies, implementing SCD Type 2 logic across 7M accounts through stakeholder requirements gathering
- Developed 12 LookML semantic models in Looker abstracting complex joins across PostgreSQL, MySQL, and MS-SQL Server, enabling self-service BI with data storytelling capabilities that reduced ad-hoc requests 50%

- Implemented event-driven architecture with Kafka consumers for real-time stream processing of 30K events/minute through microservices API integration, enabling sub-minute alerting on payment failures with data observability
- Optimized Redshift WLM queues and VACUUM operations on 15B+ event fact tables through performance tuning, reducing p95 dashboard latency from 45s to 12s via scalability improvements
- Collaborated with data science teams to build feature engineering pipelines extracting 50+ behavioral features from Hive tables using PySpark on EMR, compressing ML training data prep from 2 days to 4 hours

EDUCATION

Master of Science in Computer and Information Science

Southern Arkansas University | Aug 2021 – Dec 2022 | GPA: 3.8/4.0

Bachelor of Technology in Computer Science and Engineering

Jawaharlal Nehru Technological University | Aug 2015 – Jun 2019 | GPA: 3.0/4.0

TECHNICAL SKILLS

Programming & Scripting: Python (PySpark, Pandas, NumPy), SQL (Snowflake, Redshift, PostgreSQL, Oracle, MS-SQL Server), Java, Scala, Bash, Shell scripting, Unit testing (pytest, unittest), JSON/XML processing

Cloud Data Warehouses: Snowflake (SnowSQL, Snowpipe, Streams, Tasks, Time Travel, Zero-copy cloning), AWS Redshift, Google BigQuery, Azure Synapse

Cloud Platforms: AWS (S3, Glue, EMR, Lambda, Kinesis, Step Functions, ECS, EKS), Databricks (Delta Lake), GCP (BigQuery), Azure (Synapse, Data Factory)

Data Pipeline & Orchestration: Apache Airflow, dbt, Fivetran, Prefect, AWS Glue, Dataform, Dagster, Apache NiFi

Big Data & Streaming: Apache Spark (PySpark, Structured Streaming), Hadoop (HDFS, Hive), Kafka (Connect, Streams), Amazon Kinesis, Apache Flink

Data Lakehouse & Storage: Delta Lake, Apache Iceberg, S3, Medallion architecture, Parquet, Avro, ORC, Protocol Buffers

Data Modeling: Star/snowflake schema, Dimensional modeling, Data vault, SCD Type 1/2/3, Data pipeline architecture, Batch/stream processing, Event-driven architecture

Data Quality & Governance: Great Expectations, Deequ, dbt tests, Data validation, Data profiling, Data observability, Data lineage, Data cataloging, Master Data Management, Data contracts, Schema evolution

Databases: PostgreSQL, MySQL, Oracle, MS-SQL Server, MongoDB, DynamoDB, Redis, Query optimization, Indexing, Partitioning strategies

DevOps & Infrastructure: Docker, Kubernetes, ECS/EKS, Terraform, CloudFormation, Infrastructure as Code, GitHub Actions, Git workflows, Code review

Monitoring & BI: CloudWatch, Grafana, Tableau, Power BI, Looker, Amazon QuickSight, Troubleshooting, Incident management, Production support

Methodologies: DataOps, Agile, Scrum, Sprint planning, Stand-ups, Retrospectives, Requirements gathering, Stakeholder communication, RESTful API, CDC

Performance & Optimization: Performance tuning, Query optimization, Cost optimization, Scalability, Cloud-native architecture, Serverless computing

CERTIFICATIONS

- Microsoft Certified: Fabric Data Engineer Associate
- Snowflake SnowPro Core Certification (In Progress)
- AWS Certified Data Analytics – Specialty (In Progress)